

Slide 1 - Early Warning for Student Attrition

An Explainable Machine Learning Approach with Fairness Evaluation

Technical Presentation

Capstone Step 6

Two-stage early-warning framework:

- Enrollment
- Semester 1

Slide 2 — Data, Target & Evaluation

Dataset

UCI — Predict Students' Dropout and Academic Success

- **4,424** student records
- **36 predictors + 1 target**
- **0 missing values**
- **0 duplicate records**

Target Definition

Original classes: Dropout, Enrolled, Graduate

- **1 = Dropout**
- **0 = Enrolled + Graduate**

Enrolled students are not confirmed eventual non-dropouts, so this binary target must be interpreted carefully.

Evaluation

Primary: Dropout Recall

Supporting: Precision, F1, ROC-AUC, PR-AUC

Operational: Recall@Top10%

Slide 3 — Leakage-Safe Prediction Design

Enrollment Model

Uses **22 predictors** available at or near enrollment:

- demographics and application data
- prior academic information
- macroeconomic context

Excludes:

- Semester 1 results
- all Semester 2 variables
- Debtor
- Tuition fees up to date

The last two were excluded because their exact availability at the enrollment prediction point could not be verified.

Semester 1 Model

Uses the same **22 Enrollment predictors**, plus:

- **6 raw Semester 1 variables**
- **5 engineered features**

Total: 33 predictors

Engineered features include approval rate, completion gap, and no-approved-units indicators.

Each model only uses information available at its prediction point.

Slide 4 - Experimental Design

Leakage-Safe Evaluation

- **80/20 stratified train/test split**
- **3,539** training students
- **885** held-out test students
- `random_state = 42`

```
train_idx, test_idx = train_test_split(  
    np.arange(len(y)),  
    test_size=0.20,  
    random_state=42,  
    stratify=y  
)
```

Model Selection

Training data only:

- Stratified **5-fold cross-validation**
- Logistic Regression
- Random Forest
- Gradient Boosting
- Hyperparameter tuning
- L1 feature selection and PCA

The test set was opened only after model selection was complete.

Slide 5 — Model Selection

Tuned Logistic Regression

Stage	CV Recall	Precision	F1	ROC-AUC	PR-AUC
Enrollment	70.6%	52.4%	60.1%	76.9%	61.4%
Semester 1	77.1%	69.2%	72.9%	88.3%	82.3%

```
search = GridSearchCV(  
    estimator=pipeline,  
    param_grid=logistic_param_grid,  
    scoring=scoring,  
    refit="Recall",  
    cv=cv,  
    n_jobs=-1  
)  
  
search.fit(X_train, y_train)
```

Why Logistic Regression?

- strongest balance of **Dropout Recall** and supporting metrics
- interpretable coefficients
- simpler and more reproducible than ensemble alternatives

Hyperparameters were selected using training-set cross-validation—not the held-out test set.

Slide 6 — Held-Out Test Performance

Stage	Recall	Precision	F1	ROC-AUC	PR-AUC
Enrollment	72.2%	52.6%	60.8%	78.6%	64.5%
Semester 1	77.5%	69.8%	73.5%	89.1%	83.4%

Confusion Matrix

Stage	TP	FN	FP	TN
Enrollment	205	79	185	416
Semester 1	220	64	95	506

Semester 1 Improvement

Compared with Enrollment:

- **+15** true positives
- **-15** false negatives
- **-90** false positives

Semester 1 improves both dropout detection and targeting efficiency.

Slide 7 — Explainability: What Drives the Models?

Enrollment Model

Strongest permutation-importance signals:

- scholarship status
- course
- application mode
- gender
- admission grade
- age at enrollment

Semester 1 Model

Academic progress becomes more influential:

- curricular units approved
- Semester 1 grade
- curricular units enrolled
- approval rate
- no-approved-units indicator
- completion gap

Partial Dependence

Predicted dropout risk generally decreases as:

- approved Semester 1 units increase
- Semester 1 grades improve

The model shifts from background factors toward observed academic progress. These relationships are predictive, not causal.

Slide 8 — Fairness Audit

Fairness was evaluated on the held-out test set across:

- Gender
- Age groups
- Scholarship status

Dropout Recall Gap

Group	Enrollment	Semester 1
Gender	29.1 pts	9.7 pts
Age	45.4 pts	27.7 pts
Scholarship status	65.0 pts	40.2 pts

Equalized-Odds Gap

Group	Enrollment	Semester 1
Gender	0.340	0.108
Age	0.544	0.277
Scholarship status	0.650	0.402

Semester 1 reduces every observed subgroup gap, but meaningful disparities remain—especially for age and scholarship status.

Implication: subgroup performance must be monitored and mitigated before deployment.

Slide 9 — Limited-Capacity Intervention

Assume support is available for only the highest-risk **10% of students**.

Held-out test set: **885 students**

Intervention capacity: **89 students**

Stage	Dropouts Captured	Recall@Top10	Precision@Top10
Enrollment	73	25.7%	82.0%
Semester 1	88	31.0%	98.9%

Interpretation

Enrollment

- earlier warning
- captures 73 Dropout cases
- useful for broader, lower-cost support

Semester 1

- captures 88 Dropout cases
- almost all intervention slots correspond to actual Dropout cases
- better for higher-cost, targeted support

The model's practical value is helping prioritize limited intervention capacity.

Caution: Future-cohort validation is still required.

Slide 10 — Conclusions, Limitations & Next Steps

Conclusions

- Dropout risk can already be identified at Enrollment
- Semester 1 substantially improves predictive performance
- Later predictions rely more on observed academic progress
- Semester 1 also reduces several observed fairness gaps
- A two-stage approach provides both earlier warning and stronger later targeting

Limitations

- Single dataset and institutional context
- `Enrolled` students are included in the 0 class
- Predictive relationships are not causal
- Some fairness subgroups have limited positive-case counts
- Future-cohort performance is unknown

Before Deployment

1. Validate on future student cohorts
2. Define intervention capacity and operating thresholds
3. Test fairness-mitigation strategies
4. Monitor subgroup Recall and false-negative rates
5. Maintain human review for intervention decisions

Current Status

Promising decision-support prototype — not deployment-ready

Use the model to support student outreach, not to automate decisions about students.